LESSON PLAN

AUXILIARY EQUIPMENT (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	5	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Terminology used i.e. BDC TDC Stoke length, clearance volume, swept volume, valve lead, valve lag, ineffective crank angle
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Terminology used i.e. BDC TDC Stoke length, clearance volume, swept volume, valve lead, valve lag, ineffective crank angle
5. Specific Objectives	22.1.1.1 Trainees will be able to explain about the working principle of piston engines.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of engine terminology diagrams; Safety: Ensure no loose items on desk; Supervision: Instructor monitors diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a piston engine?
(b) Define the term 'stroke' in an engine context.
(c) List three key components found in a piston engine cylinder.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of BDC and TDC • Calculation of Stroke Length • Explanation of Clearance Volume • Explanation of Swept Volume • Relationship between volumes and displacement	L & Ds	PPT, IB	15 min
(b)	• Concept of Valve Lead and Valve Lag • Timing of Valve events relative to crank angle • Definition of Ineffective Crank Angle • Impact of timing on engine efficiency • Visualizing valve timing on cycle diagram	L & Ds	PPT, PDF	20 min
(c)	• Summary of all terminology definitions • Differentiation between swept and clearance volume • Recap of valve lead and lag effects • Review of ineffective crank angle significance • Final clarification of terms	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between BDC and TDC in your own words.
(b) How does valve lead differ from valve lag in operation?
(c) Why is the ineffective crank angle important for engine performance?

12. Home Assignment and Follow Up	Prepare a brief note linking the terminology learned in this lesson to the concepts of Efficiency, BHP, and IHP covered in the next lesson plan (LP 04).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Efficiency, BHP, IHP, FHP, SHP, knocking and detonation. Ignition system, basic principles, description and operations	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min